

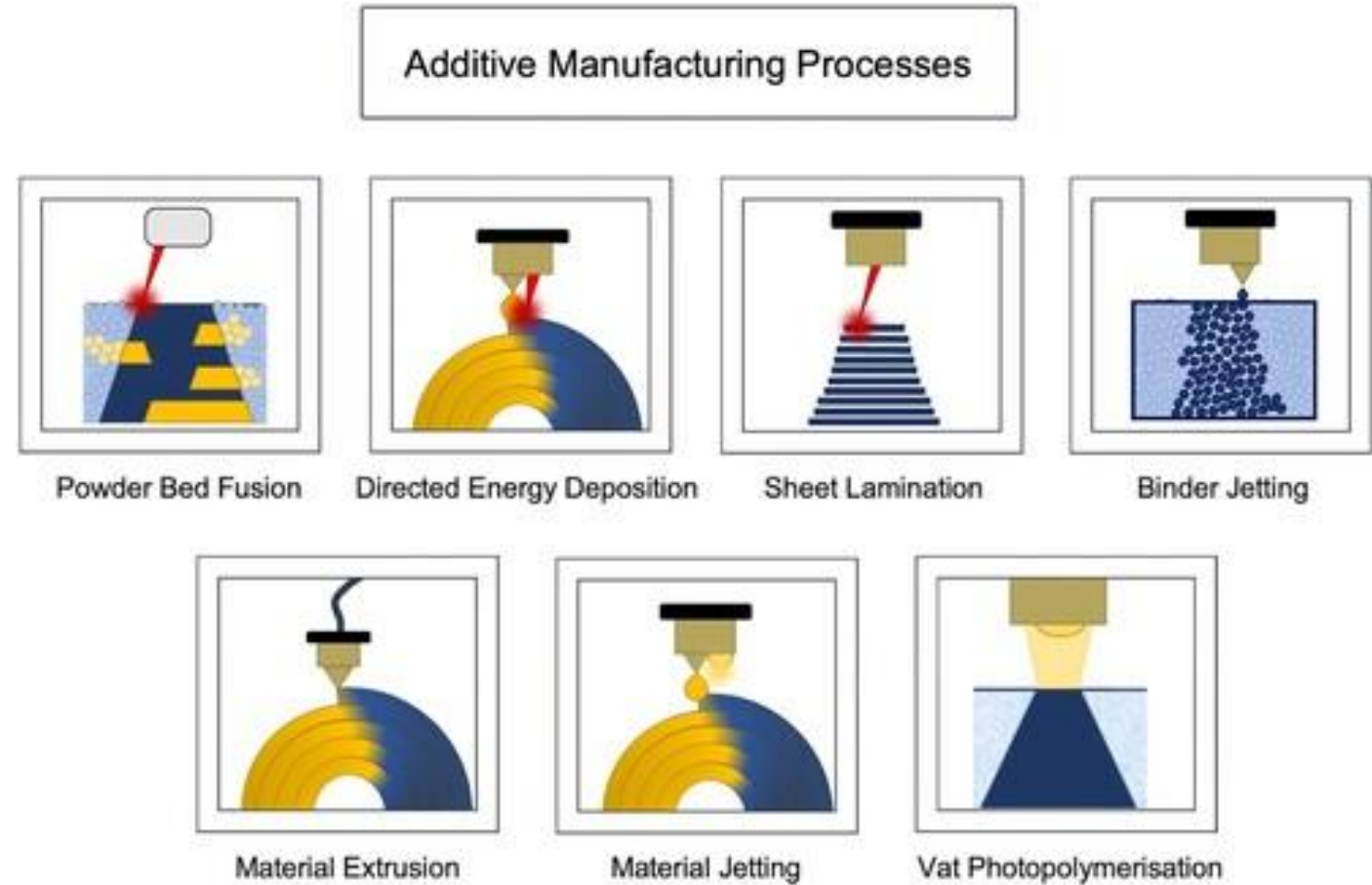
Introduction into Additive Manufacturing

MAE 1001 Guest Lecture
October 31

Sumner Gubisch

Outline

- What is additive manufacturing?
- Types of additive manufacturing
- Advantages and disadvantages of additive manufacturing
- My research in additive manufacturing
- Tips and tricks for extrusion based additive manufacturing
- Q and A

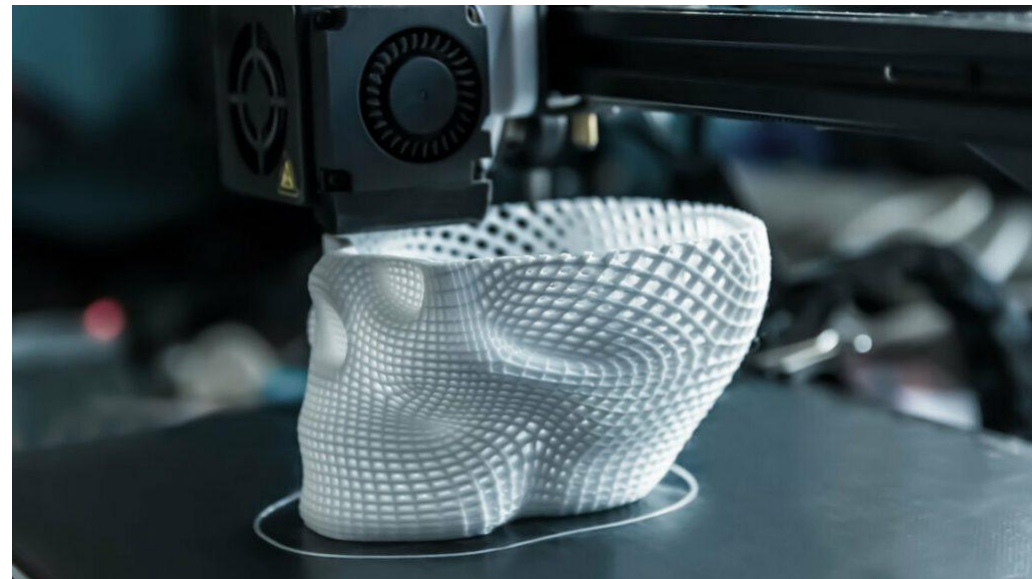


What is Additive Manufacturing?

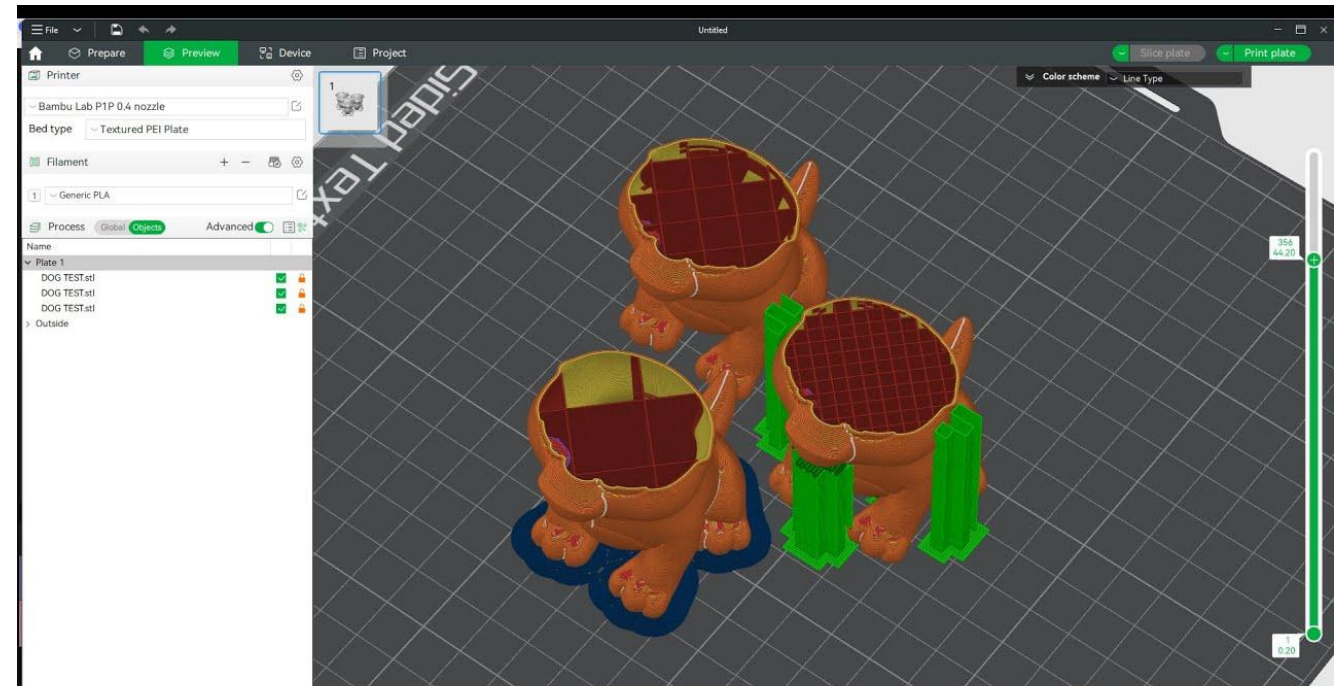
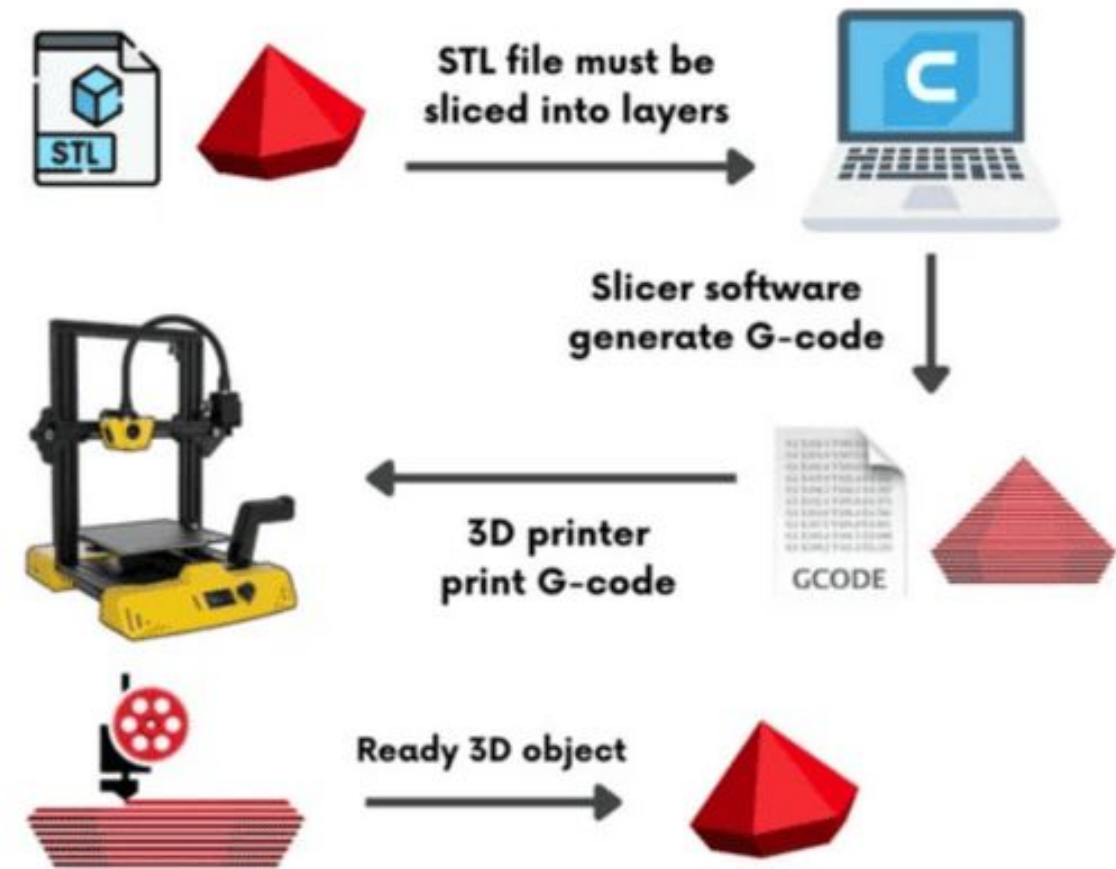
Core Definition: Additive Manufacturing, often called **3D Printing**, is a revolutionary process of creating a **three-dimensional object** by **adding material layer by layer**.

Unlike **subtractive manufacturing** (like machining), which removes material from a solid block, AM builds the object from the *bottom up*, resulting in **less waste**.

Key Benefit: This layer-by-layer approach allows for the creation of **complex geometries** and **intricate internal structures** that are impossible or too costly to produce with traditional methods.



Additive Manufacturing Slicer and G Code



Fan speed setting
Nozzle travel speed (without extrusion)
Nozzle printing speed (with extrusion)
X, Y Coordinates
Layer height
Extrusion length

```
;Layer count: 25  
;LAYER:0  
M107  
G0 F9000 X52.235 Y55.800 Z0.300  
;TYPE:SKIRT  
G1 F2340 X56.093 Y55.800 E0.18815  
G1 X56.346 Y55.605 E0.20373  
G1 X57.299 Y55.078 E0.25684  
G1 X58.540 Y54.758 E0.31934  
G1 X59.404 Y54.719 E0.36152  
G1 X60.320 Y53.688 E0.42878
```


Types of Additive Manufacturing

Extrusion Based Manufacturing (EBM)

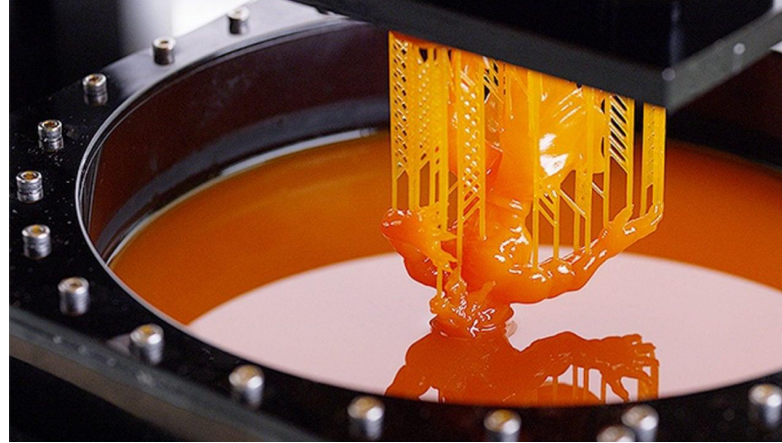


Fused Filament Fabrication (FFF)

Direct Ink Writing (DIW)

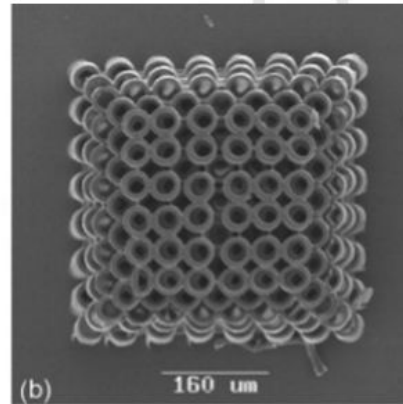


Vat Photopolymerization (VPP)

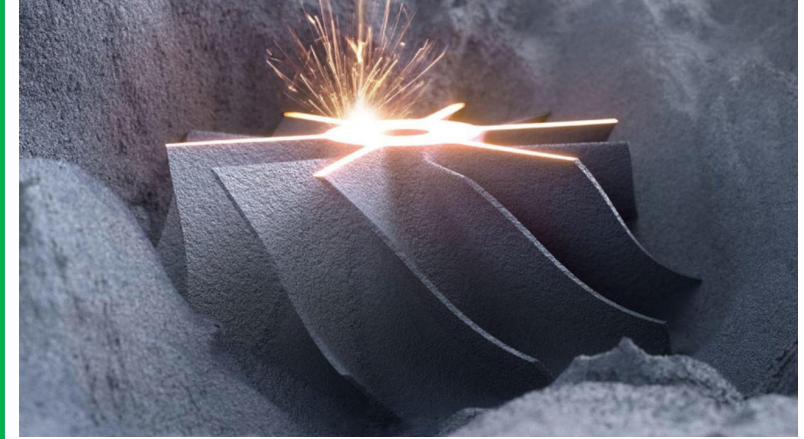


Stereolithography Apparatus (SLA)

Direct Laser Writing (DLW)



Powered Bed Fusion



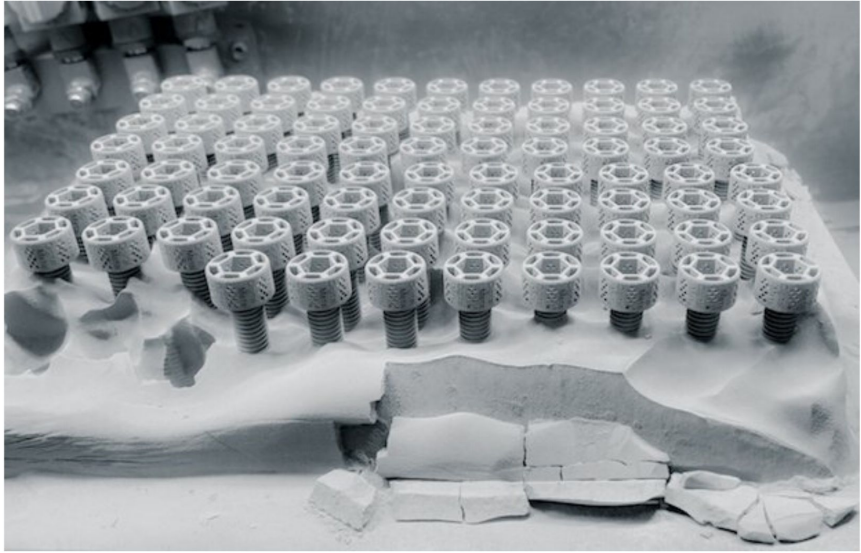
Laser Powder Bed Fusion (LPBF)

Direct Energy Deposition (DED)



Other Types of Additive Manufacturing

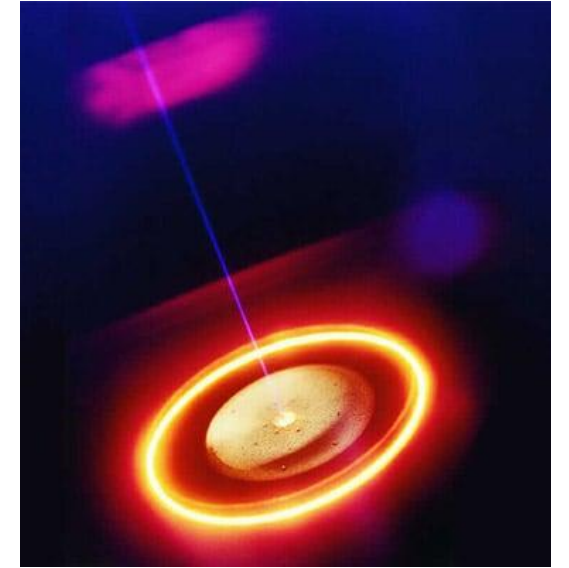
Binder Jet Printing (BJP)



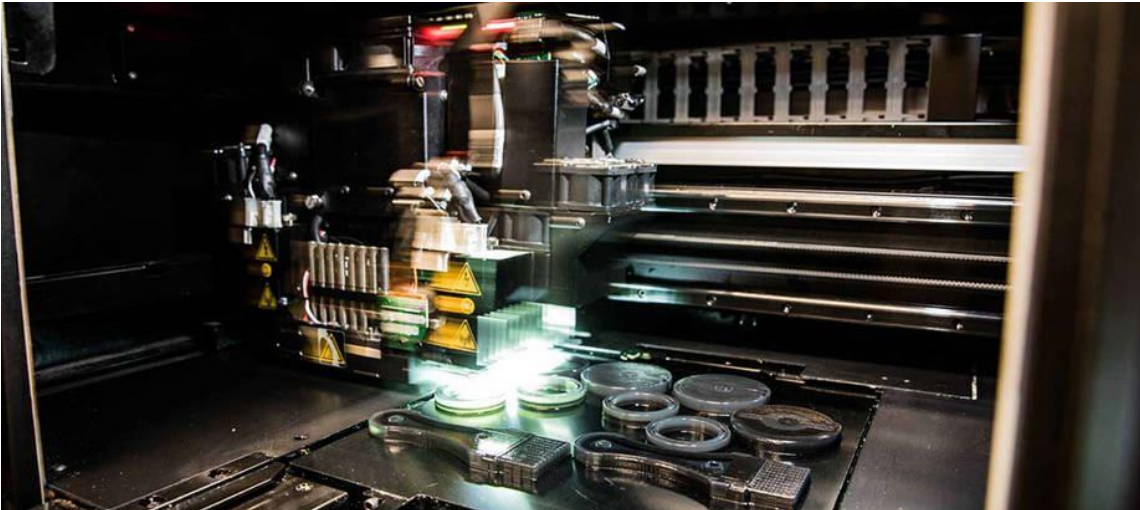
Selective Laser Sintering (SLS)



Electron Beam Melting (EBM)



Poly Jet / Ink Jet printing (IJP)



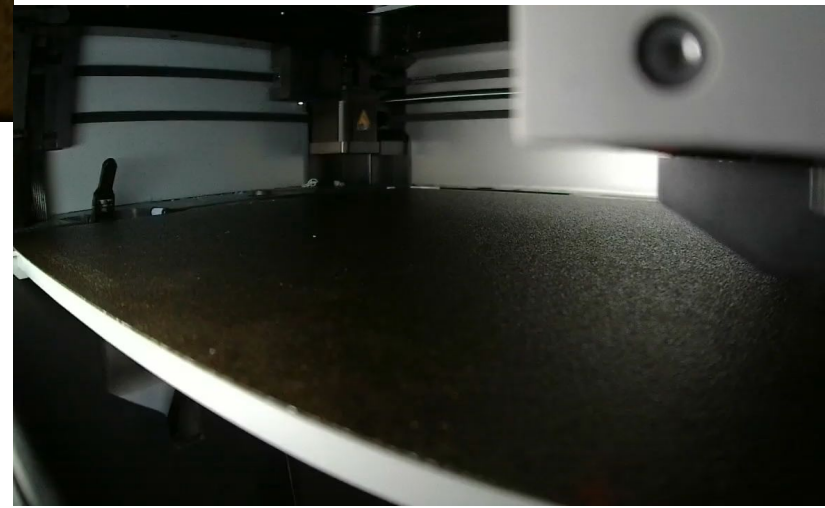
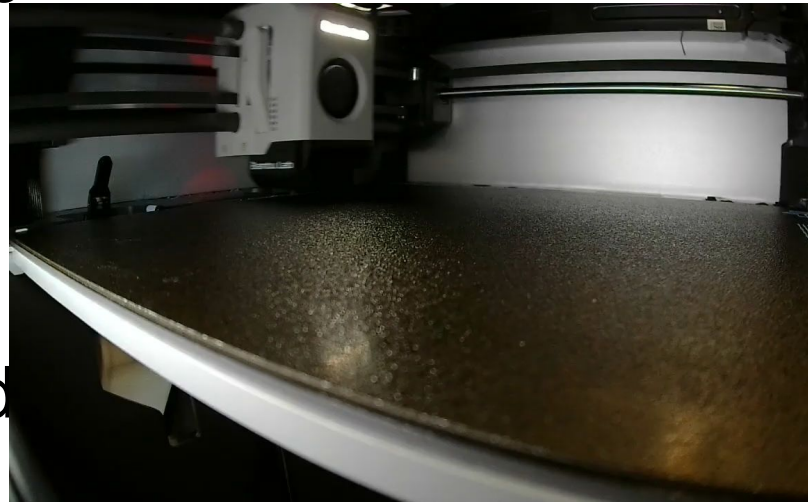
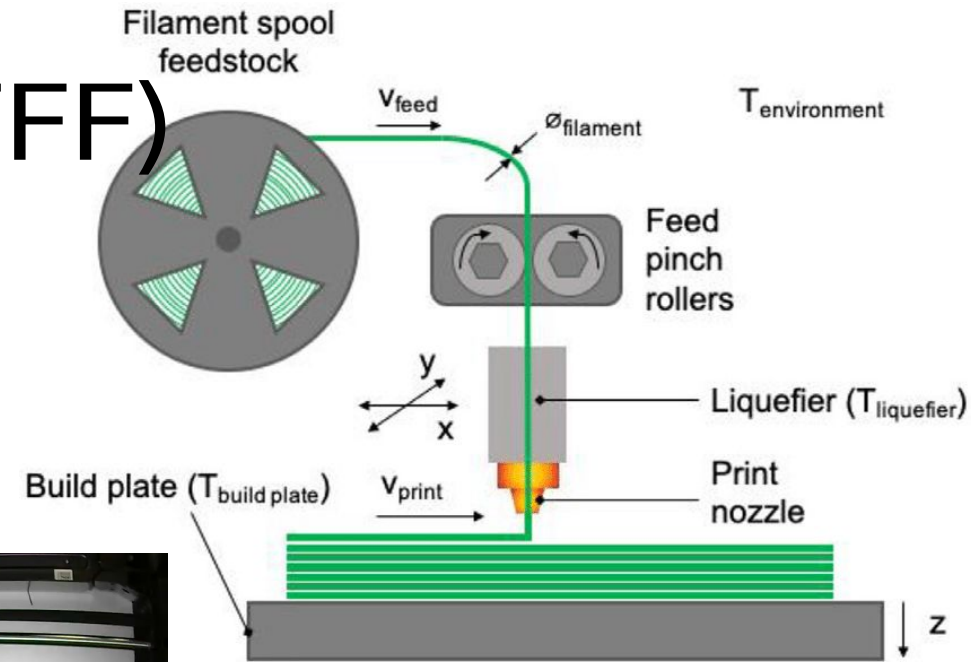
Electron Beam Additive Manufacturing (EBAM)



Fused Filament Fabrication (FFF)

Processing Steps

1. Thermoplastic is fed through a heated nozzle
2. The melted thermoplastic is pushed through the nozzle with the unmelted filament
3. The nozzle moves in x and y directions, and bed moves in z
4. Filament solidifies and a solid part is produced
5. Supports are removed as needed



Fused Filament Fabrication Post Processing

Standard Support



Standard Support

Tree Support

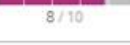
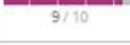
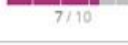
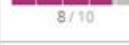
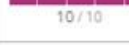
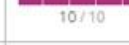
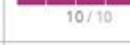
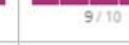
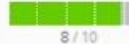
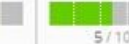
Dissolvable
Support



PVA / HIPS

Can dissolve in water or IPA

Fused Filament Materials

													
	ABS	Flexible	PLA	HIPS	PETG	Nylon	Carbon Fiber Filled	ASA	Polycarbonate	Polypropylene	Metal Filled	Wood Filled	PVA
	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More	Learn More
Compare Selected	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Ultimate Strength	 40 MPa	 26 - 43 MPa	 65 MPa	 32 MPa	 53 MPa	 40 - 85 MPa	 45 - 48 MPa	 55 MPa	 72 MPa	 32 MPa	 20 - 30 MPa	 46 MPa	 78 MPa
Stiffness	 5 / 10	 1 / 10	 7.5 / 10	 10 / 10	 5 / 10	 5 / 10	 10 / 10	 5 / 10	 6 / 10	 4 / 10	 10 / 10	 8 / 10	 3 / 10
Durability	 8 / 10	 9 / 10	 4 / 10	 7 / 10	 8 / 10	 10 / 10	 3 / 10	 10 / 10	 10 / 10	 9 / 10	 4 / 10	 3 / 10	 7 / 10
Maximum Service Temperature	98 °C	60 - 74 °C	52 °C	100 °C	73 °C	80 - 95 °C	52 °C	95 °C	121 °C	100 °C	52 °C	52 °C	75 °C
Coefficient of Thermal Expansion	90 $\mu\text{m/m}^\circ\text{C}$	157 $\mu\text{m/m}^\circ\text{C}$	68 $\mu\text{m/m}^\circ\text{C}$	80 $\mu\text{m/m}^\circ\text{C}$	60 $\mu\text{m/m}^\circ\text{C}$	95 $\mu\text{m/m}^\circ\text{C}$	57.5 $\mu\text{m/m}^\circ\text{C}$	98 $\mu\text{m/m}^\circ\text{C}$	69 $\mu\text{m/m}^\circ\text{C}$	150 $\mu\text{m/m}^\circ\text{C}$	33.75 $\mu\text{m/m}^\circ\text{C}$	30.5 $\mu\text{m/m}^\circ\text{C}$	85 $\mu\text{m/m}^\circ\text{C}$
Density	1.04 g/cm ³	1.19 - 1.23 g/cm ³	1.24 g/cm ³	1.03 - 1.04 g/cm ³	1.23 g/cm ³	1.06 - 1.14 g/cm ³	1.3 g/cm ³	1.07 g/cm ³	1.2 g/cm ³	0.9 g/cm ³	2 - 4 g/cm ³	1.15 - 1.25 g/cm ³	1.23 g/cm ³
Price (per kg)	\$10 - \$40	\$30 - \$70	\$10 - \$40	\$24 - \$32	\$20 - \$60	\$25 - \$65	\$30 - \$80	\$38 - \$40	\$40 - \$75	\$60 - \$120	\$50 - \$120	\$25 - \$55	\$40 - \$110
Printability	 8 / 10	 6 / 10	 9 / 10	 6 / 10	 9 / 10	 8 / 10	 8 / 10	 7 / 10	 6 / 10	 4 / 10	 7 / 10	 8 / 10	 5 / 10
Extruder Temperature	220 - 250 °C	225 - 245 °C	190 - 220 °C	230 - 245 °C	230 - 250 °C	220 - 270 °C	200 - 230 °C	235 - 255 °C	260 - 310 °C	220 - 250 °C	190 - 220 °C	190 - 220 °C	185 - 200 °C
Bed temperature	95 - 110 °C	45 - 60 °C	45 - 60 °C	100 - 115 °C	75 - 90 °C	70 - 90 °C	45 - 60 °C	90 - 110 °C	80 - 120 °C	85 - 100 °C	45 - 60 °C	45 - 60 °C	45 - 60 °C
Heated Bed	Required	Optional	Optional	Required	Required	Required	Optional	Required	Required	Required	Optional	Optional	Required
Recommended Build Surfaces	Kapton Tape, ABS Slurry	PEI, Painter's Tape	Painter's Tape, Glue Stick, Glass Plate, PEI	Glass Plate, Glue Stick, Kapton Tape	Glue Stick, Painter's Tape	Glue Stick, PEI	Painter's Tape, Glue Stick, Glass Plate, PEI	Glue Stick, PEI	PEI, Commercial Adhesive, Glue Stick	Packing Tape, Polypropylene Sheet	Painter's Tape, Glue Stick, PEI	Painter's Tape, Glue Stick, PEI	PEI, Painter's Tape
Other Hardware Requirements	Heated Bed, Enclosure Recommended	Part Cooling Fan	Part Cooling Fan	Heated Bed, Enclosure Recommended	Heated Bed, Part Cooling Fan	Heated Bed, Enclosure Recommended, May Require All Metal Hotend	Part Cooling Fan	Heated Bed	Heated Bed, Enclosure Recommended, All Metal Hotend	Heated Bed, Enclosure Recommended, Part Cooling Fan	Wear Resistant or Stainless Steel Nozzle, Part Cooling Fan	Part Cooling Fan	Heated Bed, Part Cooling Fan

Stereolithography Apparatus (SLA)

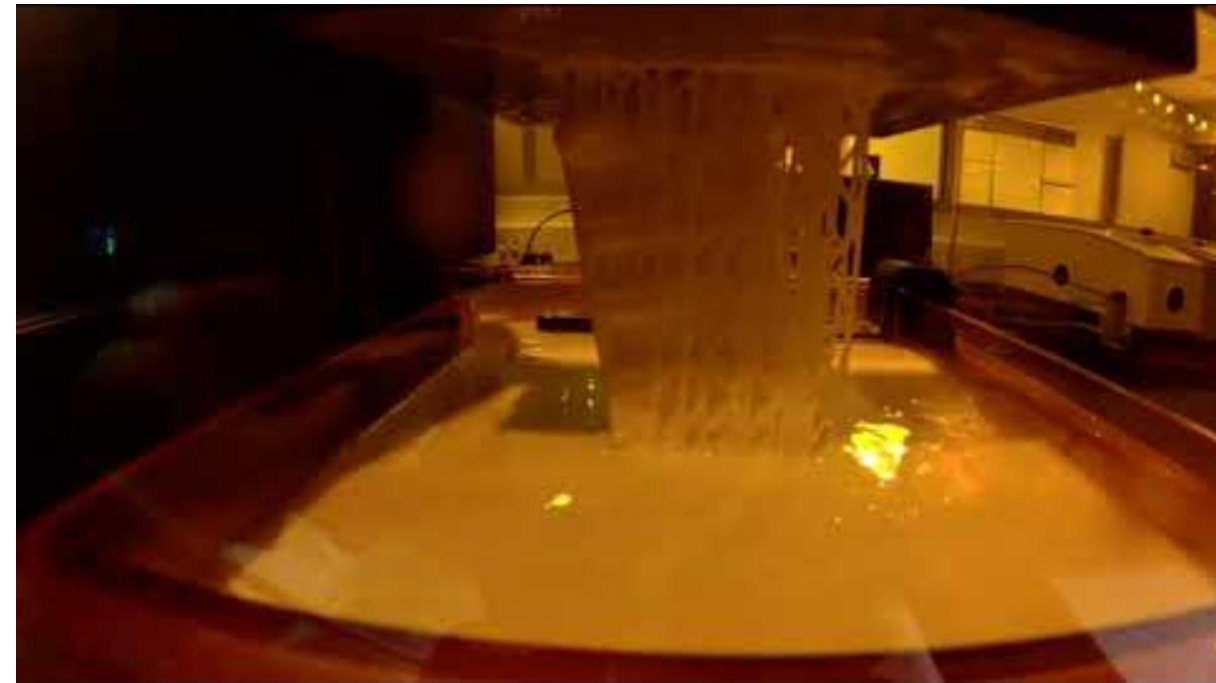
Processing Steps

1. Make sure resin levels are appropriate
2. UV laser light is used to cure the resin to the build plate
3. Laser moves back and forth and cure the resin
4. Build plate moves up to allow new resin in
5. Layer cured again etc.
6. Parts cleaned
7. Parts cured
8. Supports removed as needed



Upside-Down (Inverted) SLA

- 1 Printed Part
- 2 Supports
- 3 Resin
- 4 Build Platform
- 5 UV Laser
- 6 Galvanometers
- 7 X-Y Scanning Mirror
- 8 Laser Beam
- 9 Resin Tank



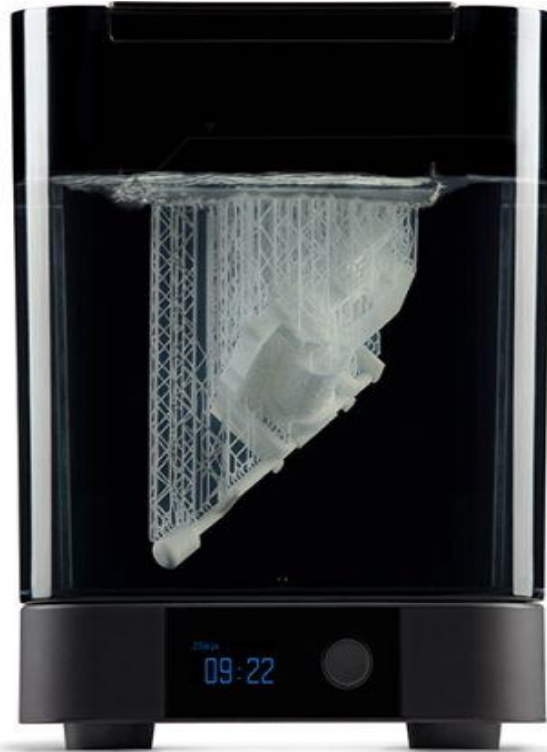
Stereolithography Apparatus Post Processing

1.



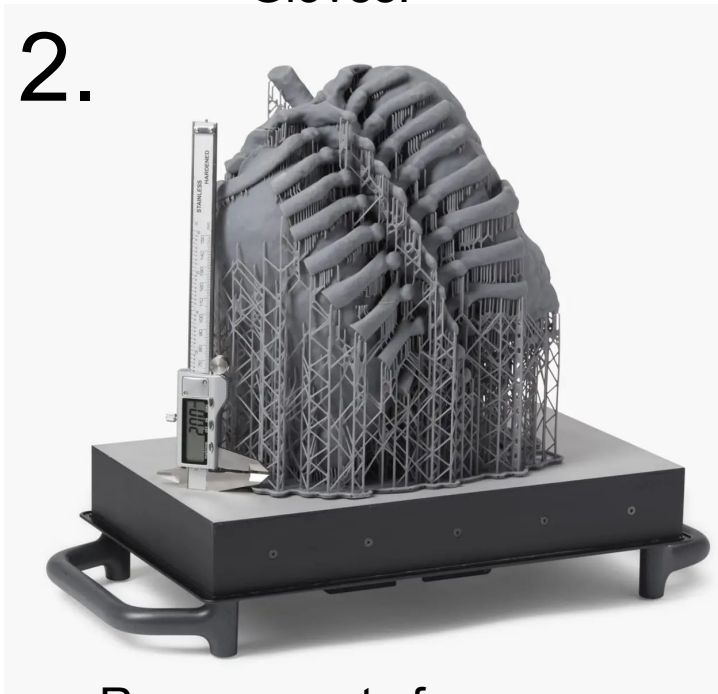
Gloves!

3.



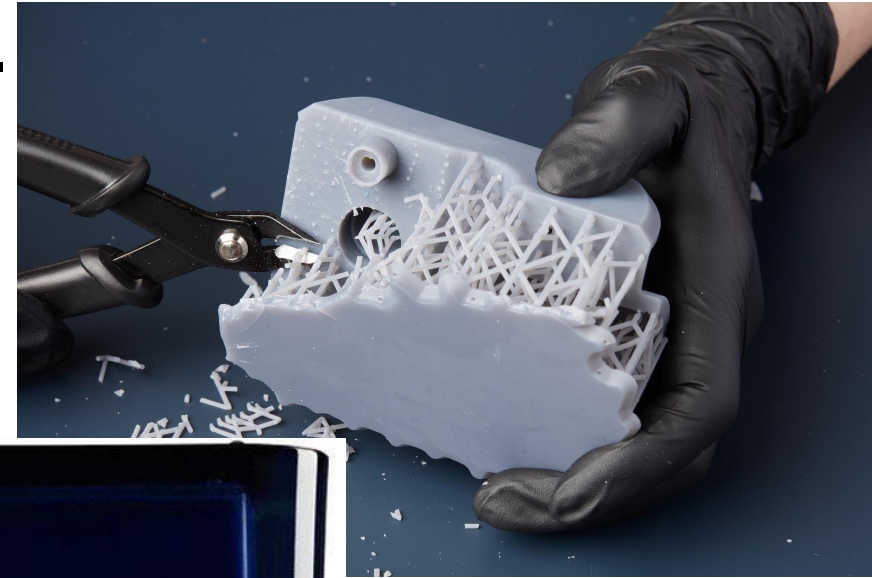
Washing in IPA or
Material Specific
Wash

2.



Remove parts from

5.



Final Support
Removal

4.



Additional Curing

Stereolithography Materials

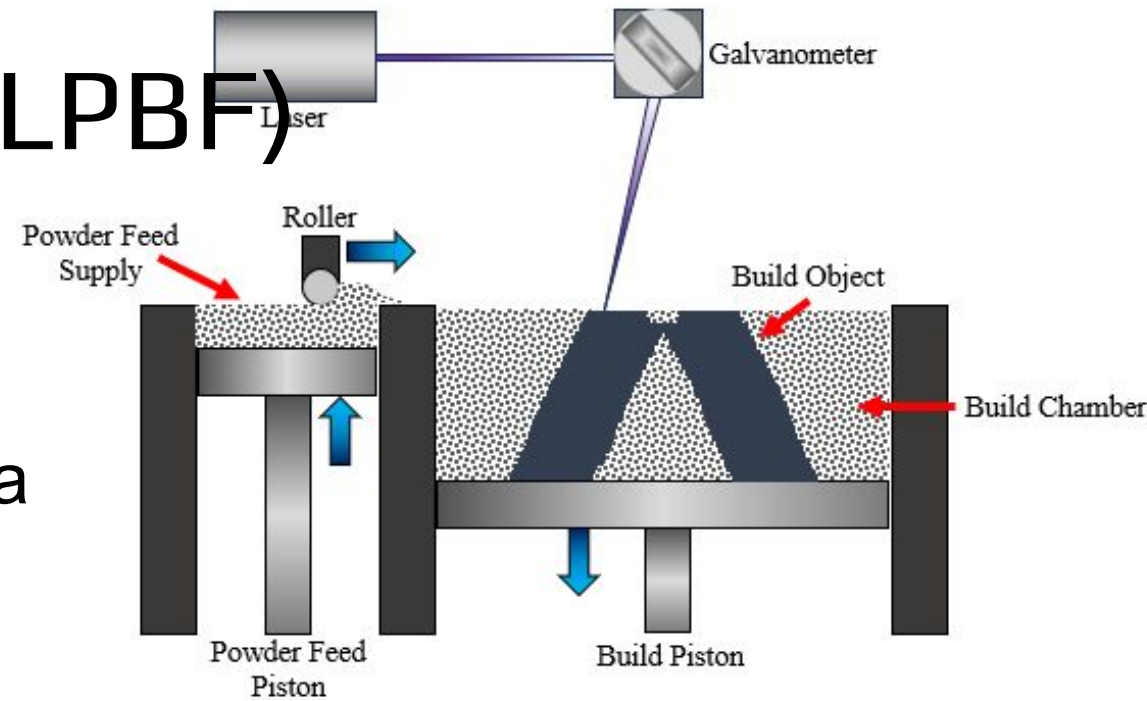


Smaller material portfolio but not nothing;
can do a lot of interesting things

Laser Powder Bed Fusion (LPBF)

Processing Steps

1. Powder is spread across the build chamber
2. A laser rasters across the build plate in a specific pattern
3. The laser then melts the powder into a specific pattern
4. Build platform lowers
5. New powder is spread over the part ect.
6. Excess powder is vacuumed up
7. Part is cut off of build plate
8. Supports are then removed
9. Final part can be polished

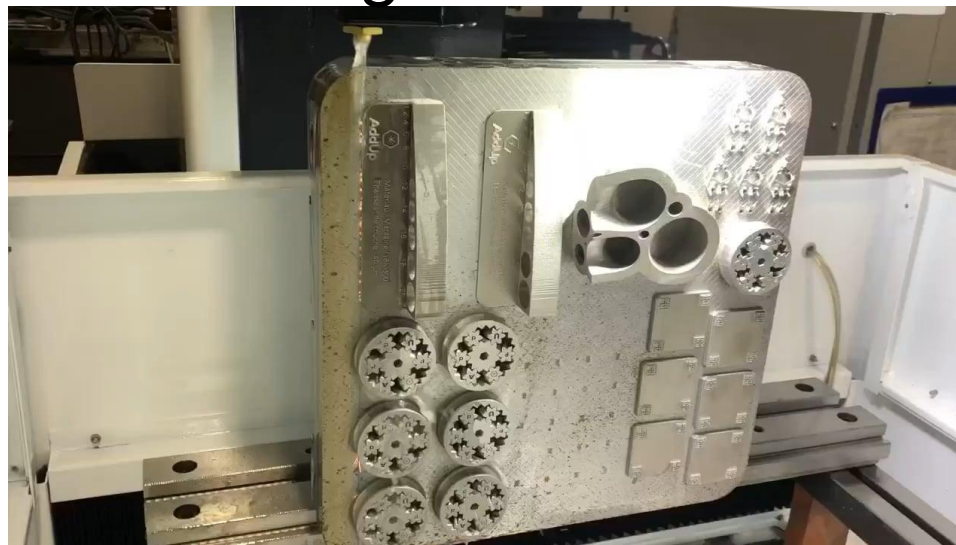


Laser Powder Bed Fusion Post Processing

1. Respirator!



3. Slicing off Build Plate



5. Polishing



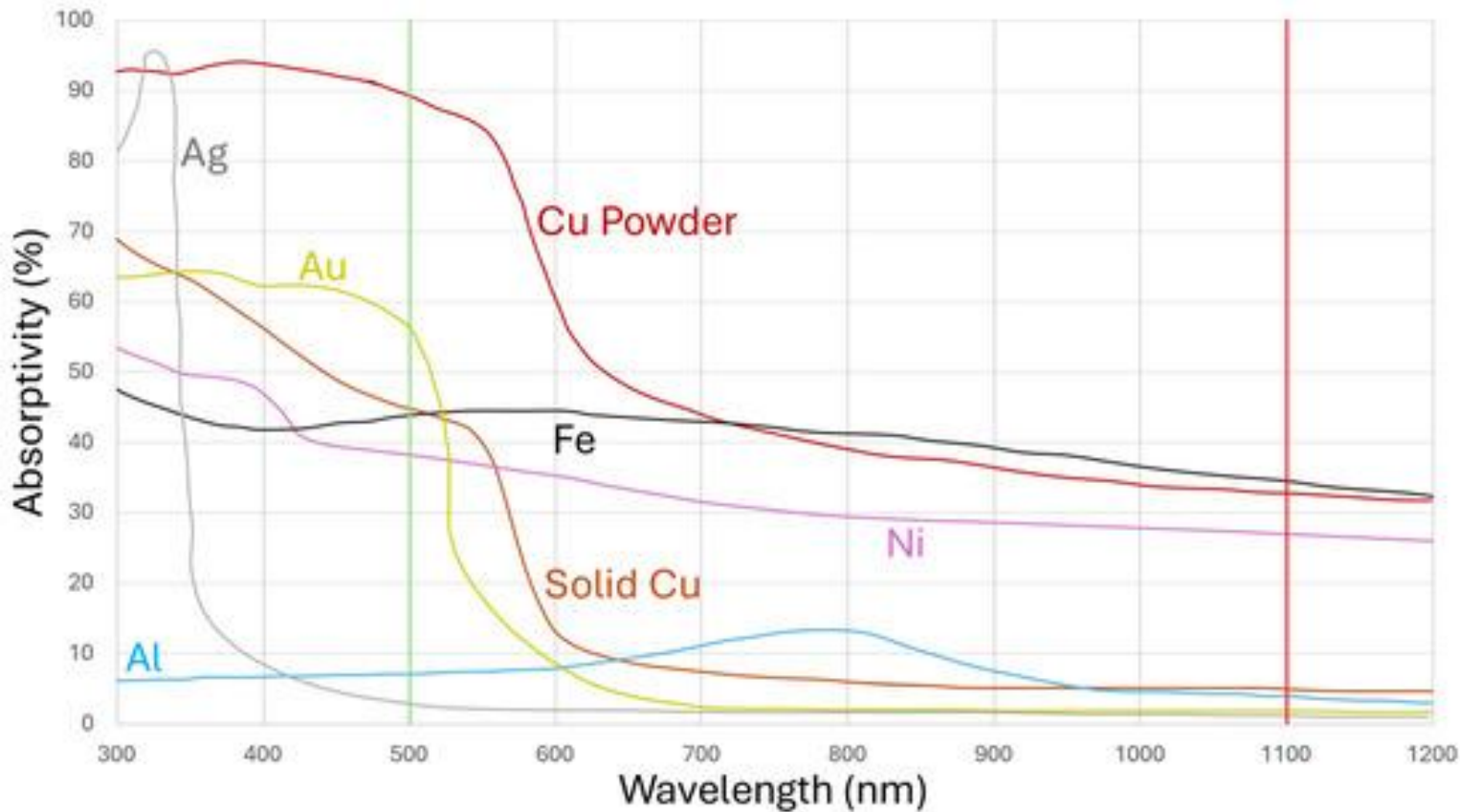
2. Powder Removal



Support Removal

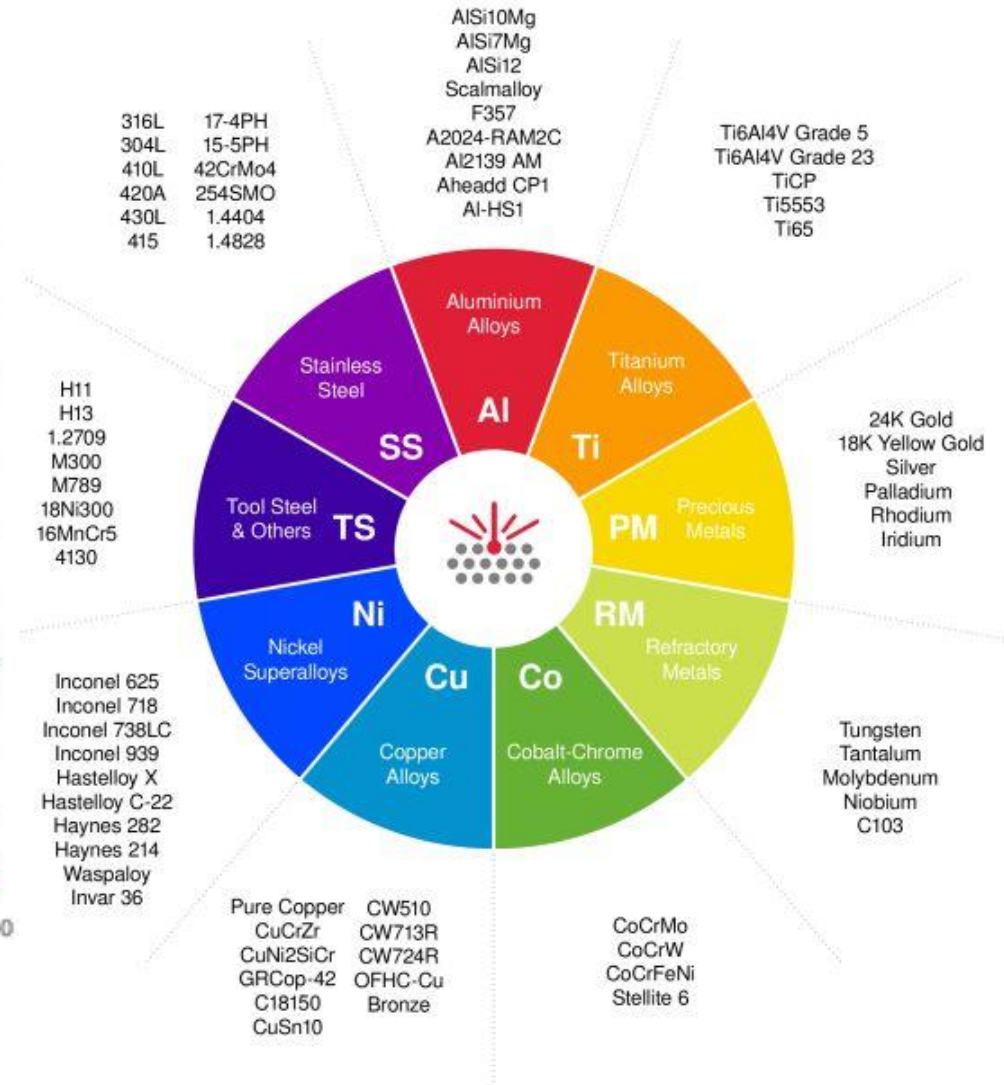


Laser Powder Bed Fusion Materials



LPBF Material Universe

Covering Commercial Alloys as well as R&D Materials



The material list presented above is compiled from publicly available sources and manufacturer data sheets as of the current publication date. It includes both commercially released and research stage materials claimed to be compatible with Laser Powder Bed Fusion (LPBF) technology. The accuracy, completeness, or applicability of the information is not warranted.

Advantages vs Disadvantages of Additive Manufacturing

Advantages

- Can print complex designs
- Consolidation of multiple parts
- Mass customizability
- Single Tooling cost
- Reduced material waste
 - Only put material where it is needed
- Locally producing parts
- Rapid prototyping

Disadvantages

- Anisotropy for many parts being produced.
- Limited part size
- Reduced surface finished
- Cost per item made doesn't change much
- Slow production speed
- High material cost and equipment cost
- Post processing cost

When to Use Additive Manufacturing

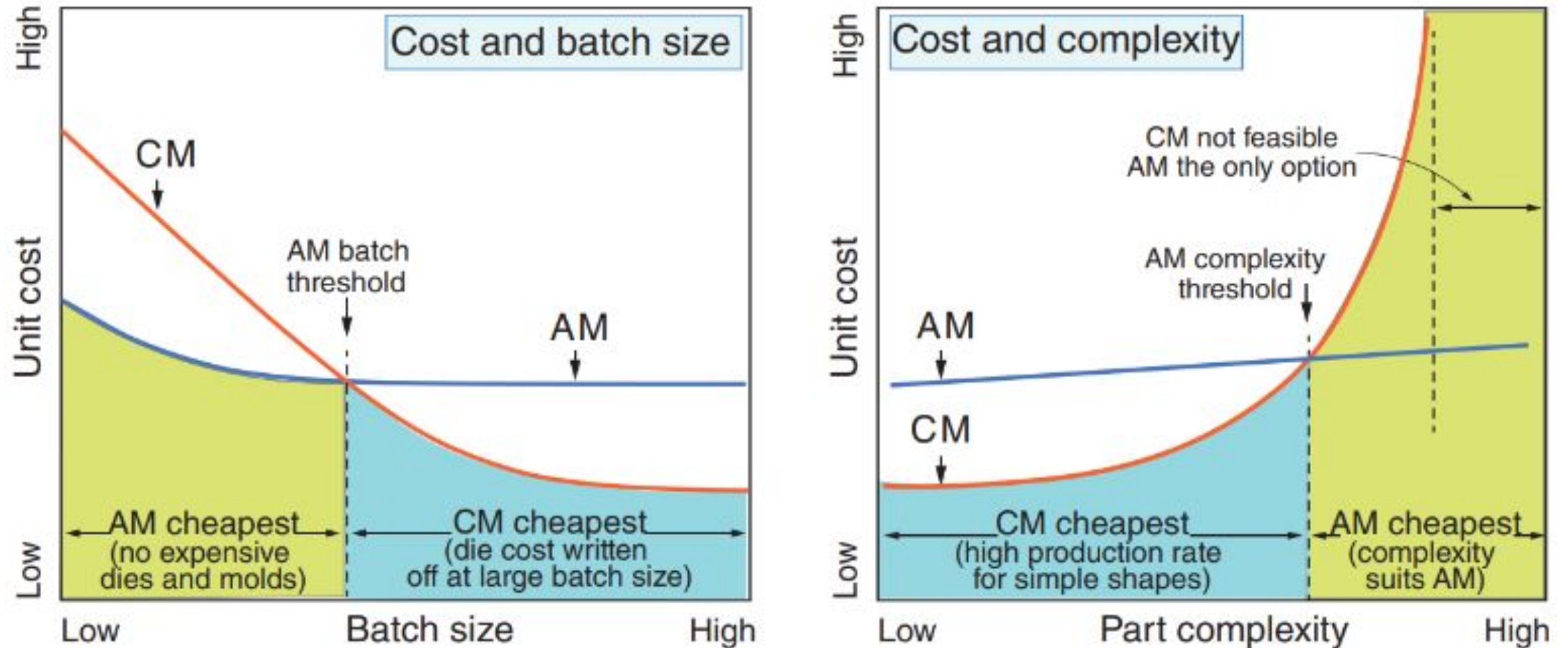
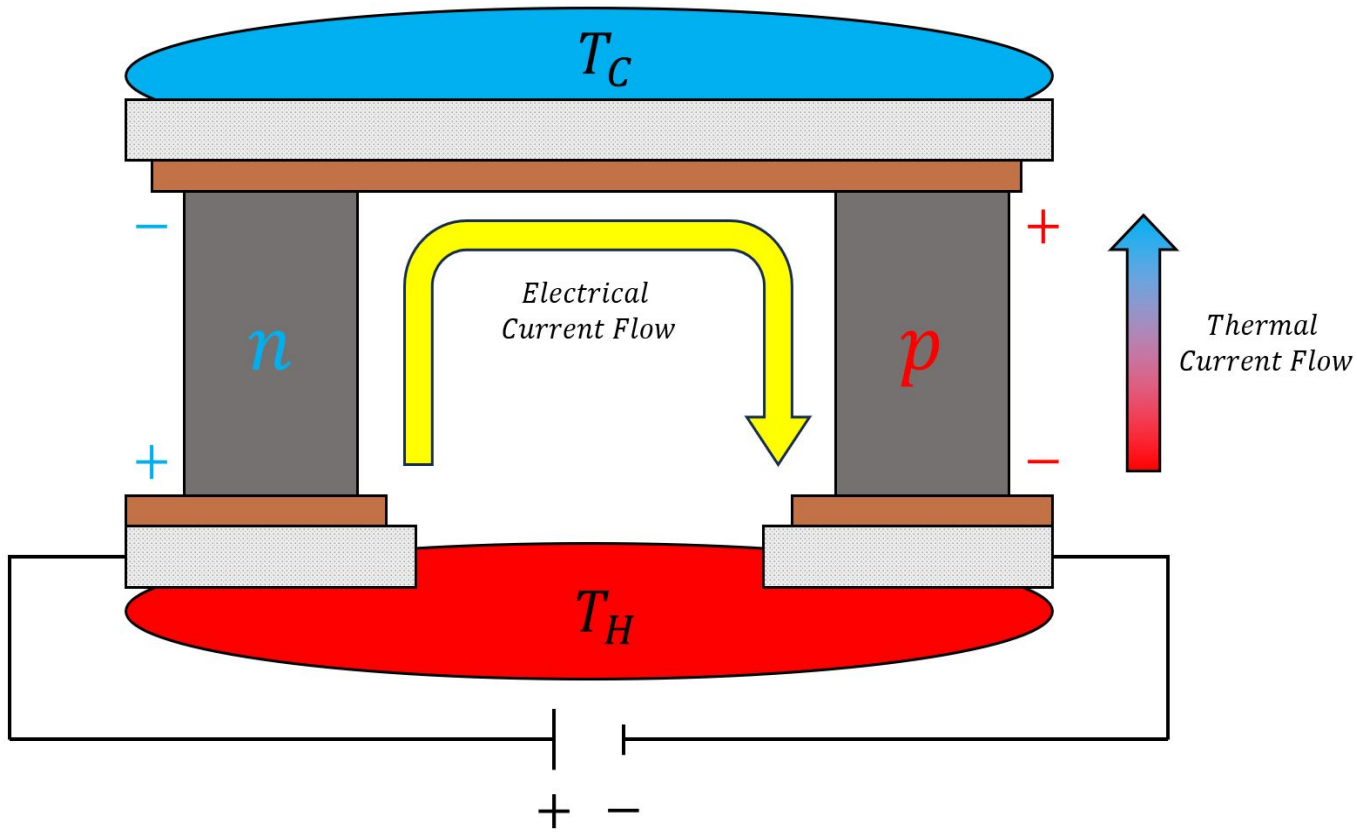


FIGURE 8.6

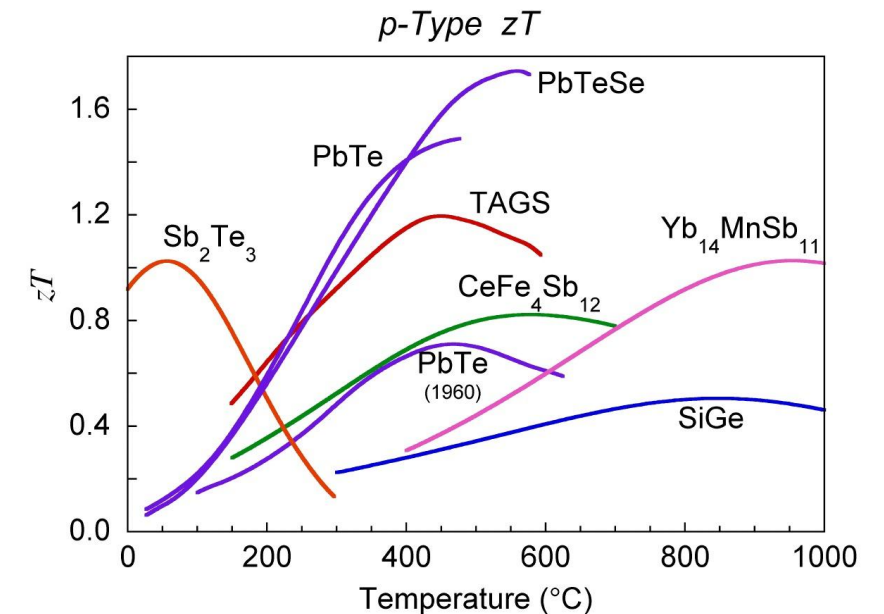
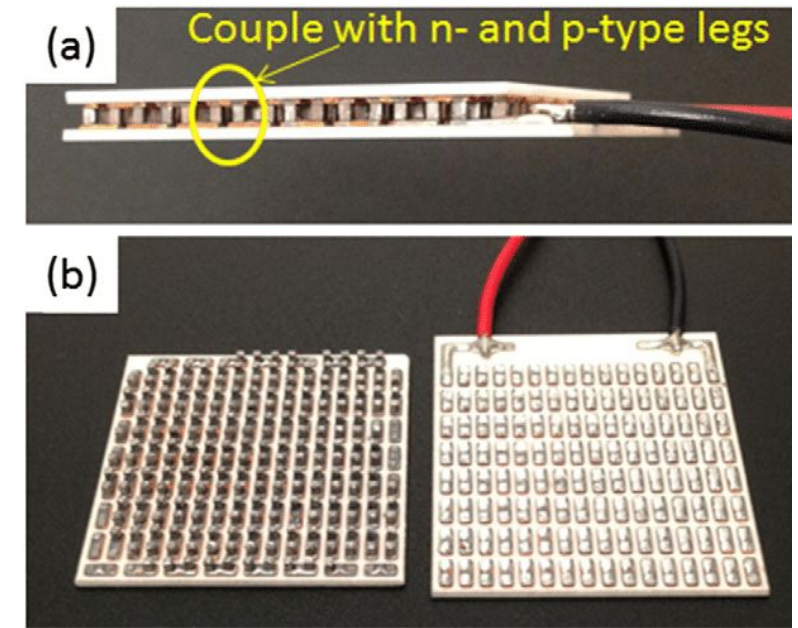
Batch size and part complexity determine the relative costs of AM and conventionally manufactured (CM) parts. (After Leary, M, 2019. *Design for Additive Manufacturing*, Elsevier.).

Introduction to Thermoelectrics



$$zT = \frac{S^2 \sigma}{\kappa} T$$

S = Seebeck coefficient
 σ = Electrical conductivity
 κ = Thermal conductivity
 T = Temperature



Motivation of Thermoelectrics

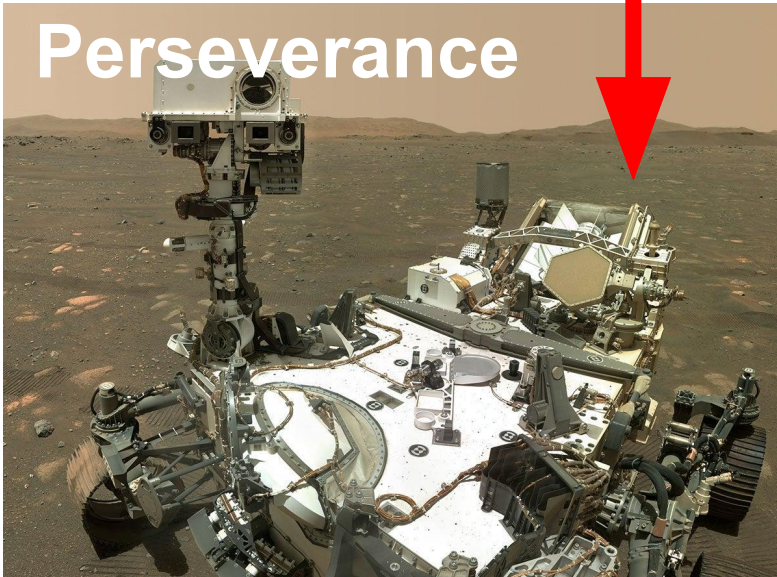
New Horizons



RTG

RTG

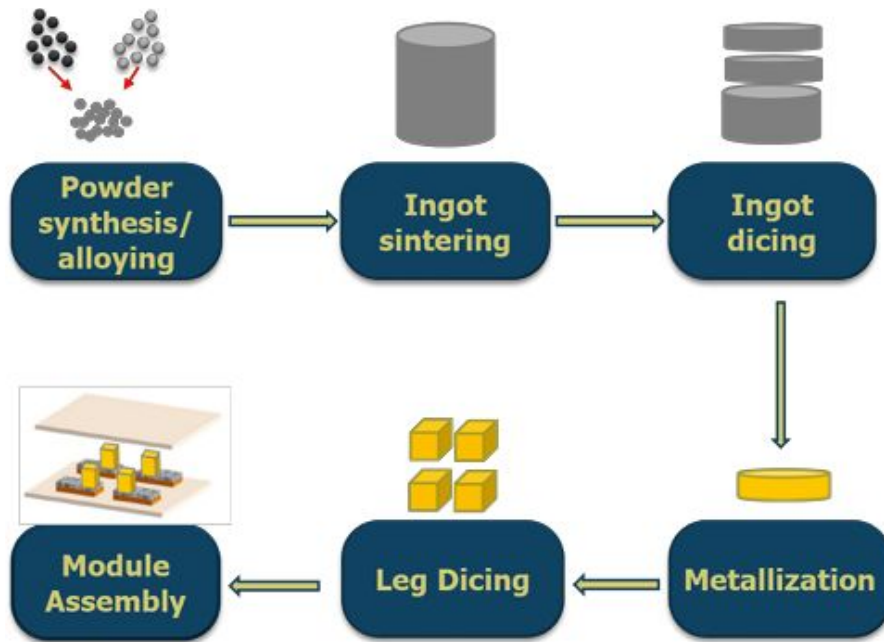
Perseverance



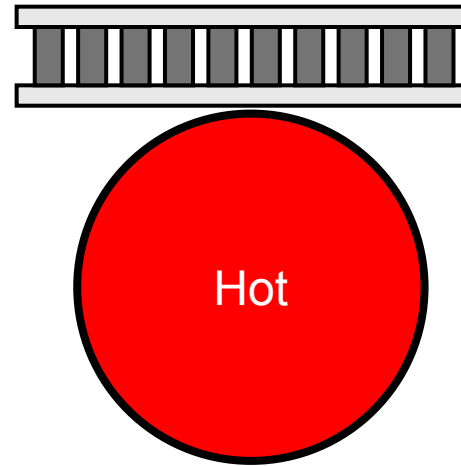
**Artist's rendering of
hypersonic jet**

Limitations of Thermoelectrics

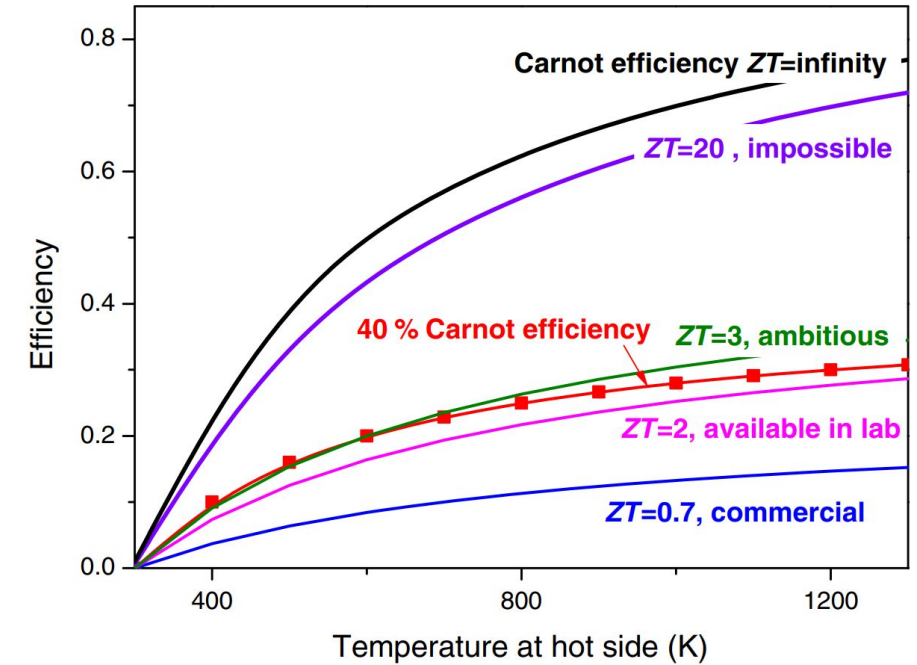
Complex, expensive, 50% loss of original material



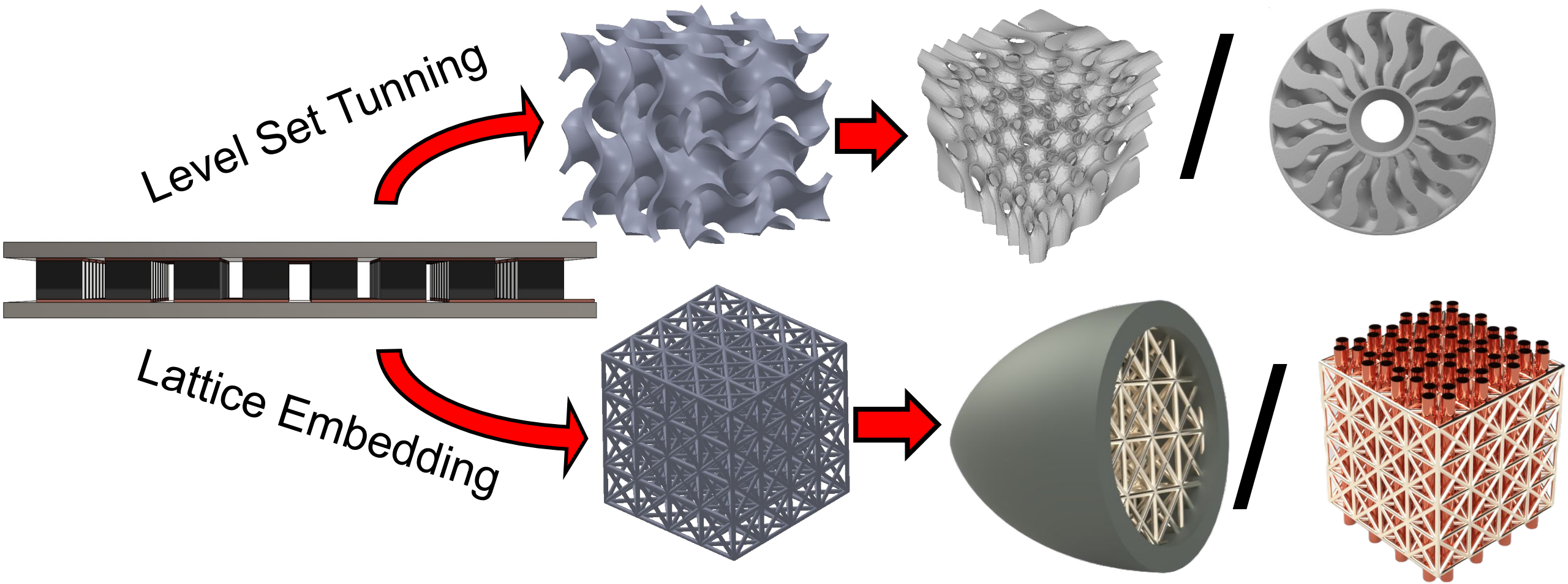
Traditional manufacturing limits to planar geometries



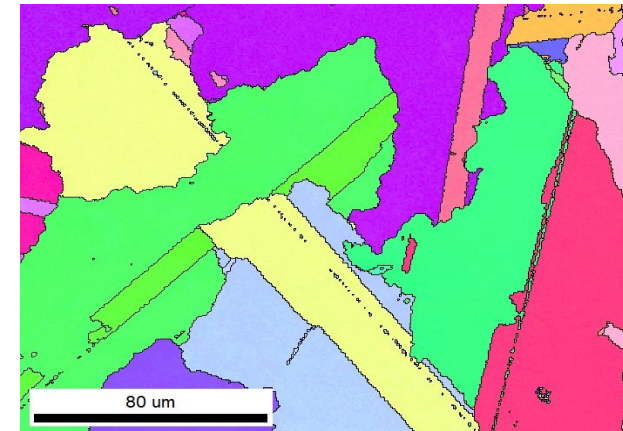
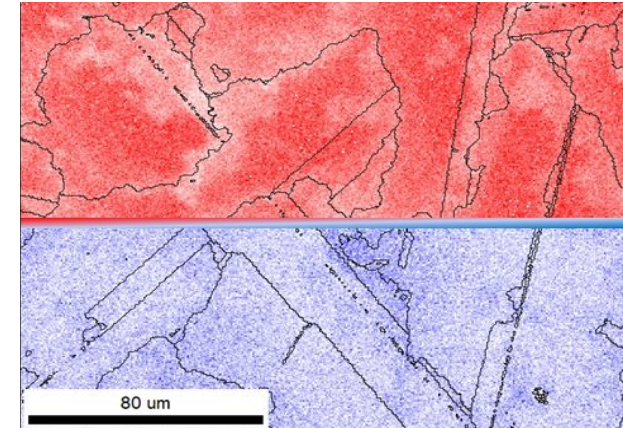
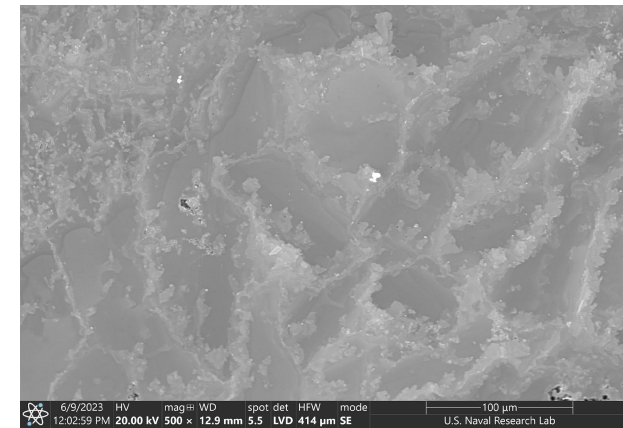
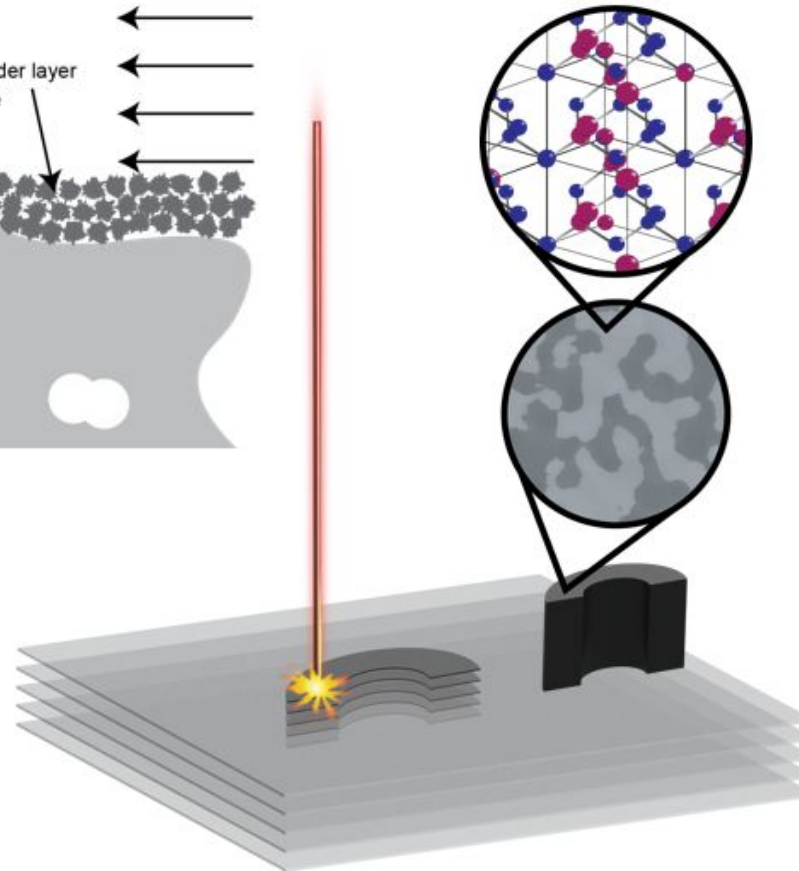
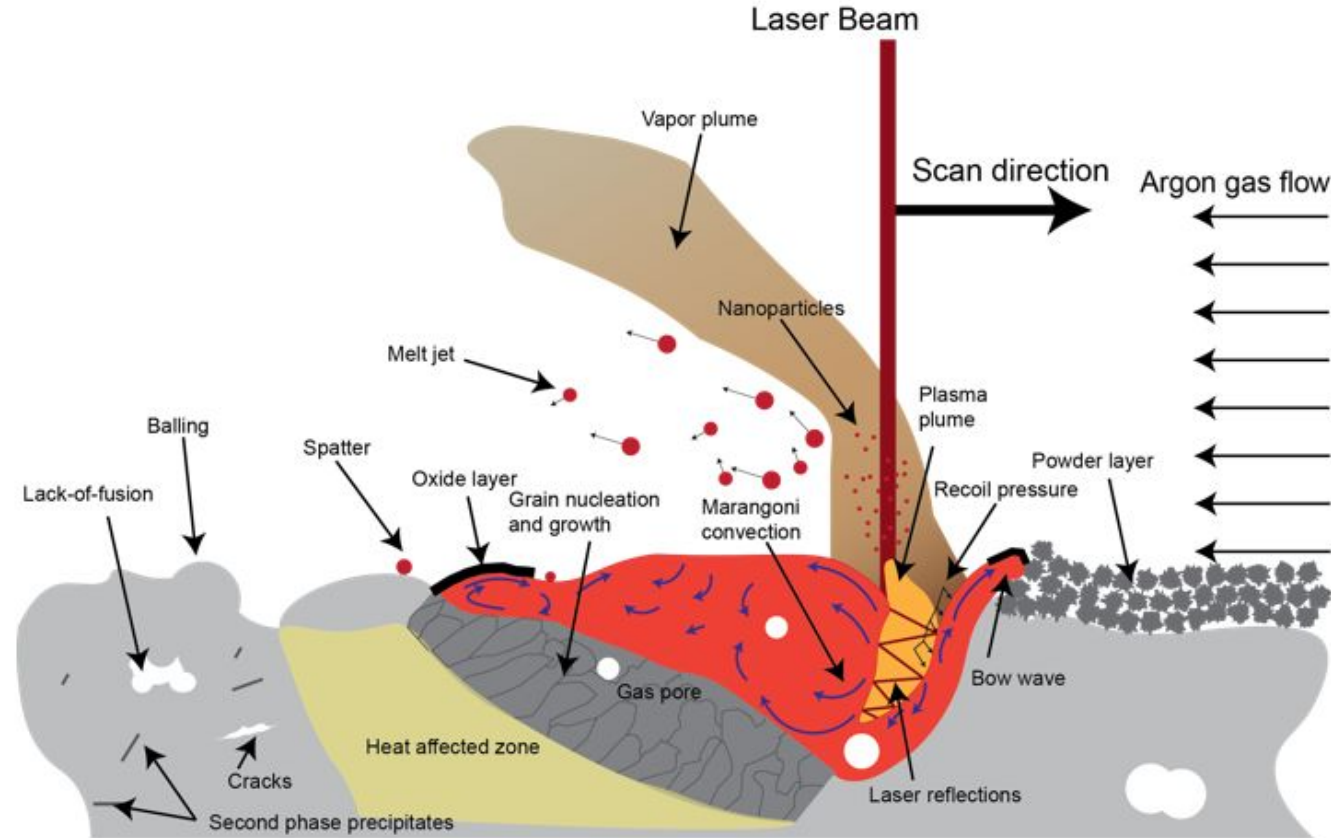
Low efficiency



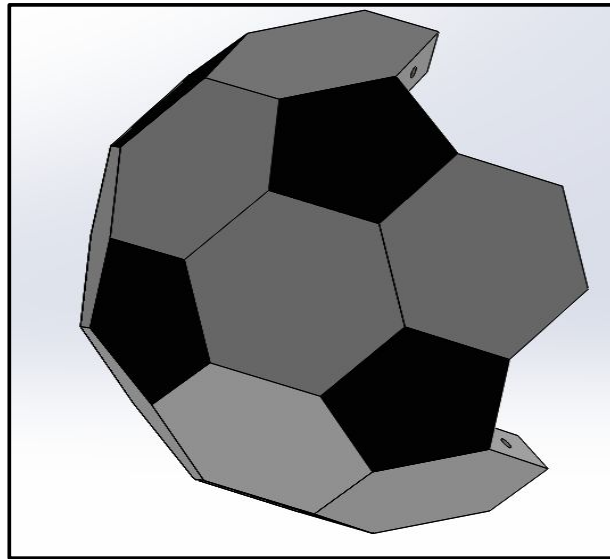
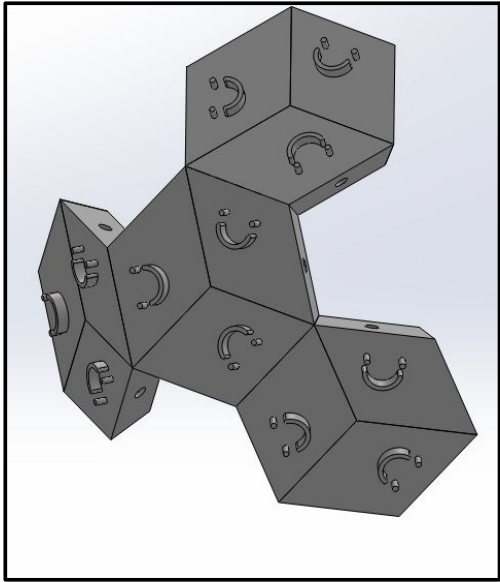
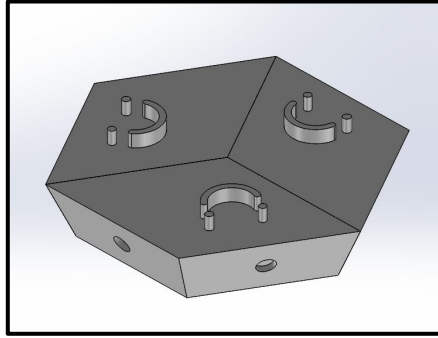
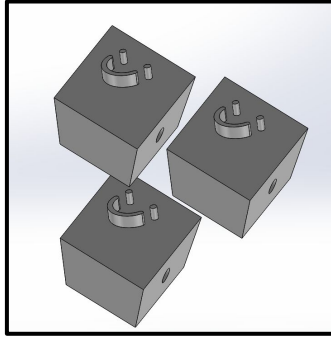
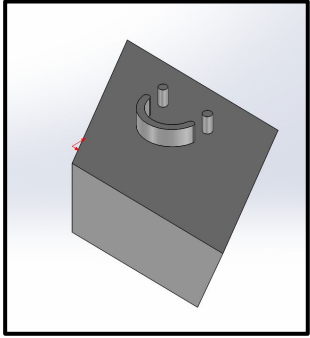
Change the Macrostructure



Change the Microstructure



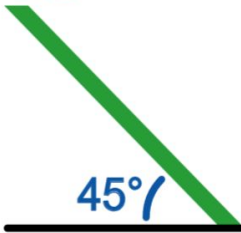
Soccer Ball Project – Tile Prints



FFF Printing Tips

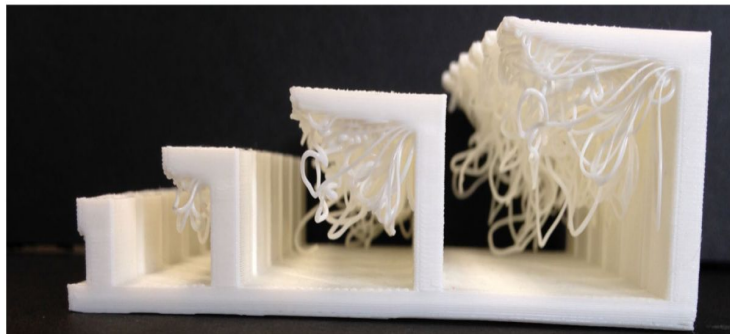
Support Considerations

- > 45° support recommended
- More than 45° with extra material
- Manually designed support .4m gap horizontally 1mm gap vertically



Overhangs

- Sensitive to overhangs
- > 1cm results in



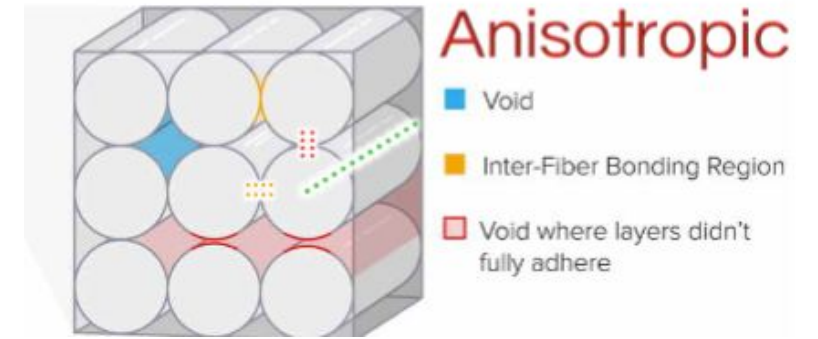
Tolerance

- 0.1 mm vertically
- 0.2 mm horizontally
- 0.5 mm for engraved
- 0.9 mm for embossed
- 1.8 mm pins
- 2mm for holes



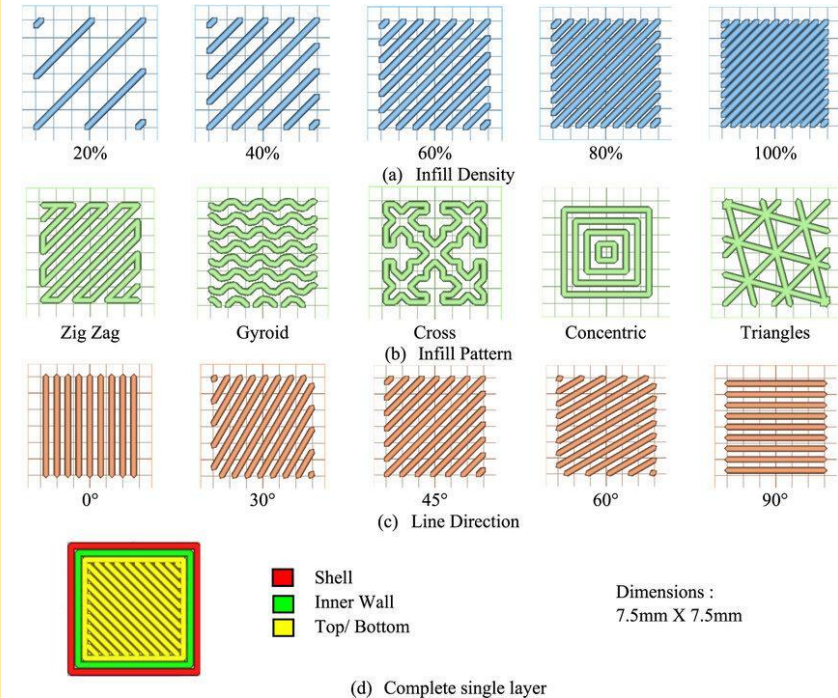
Bridging

- Can be a challenge >10 cm
- As it cools it can contract and pull the bridge upwards



Infill

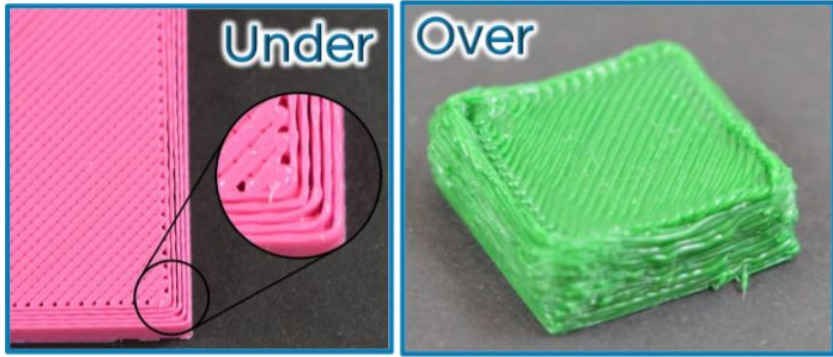
- Infill important for strength
- Number of shells matter more



FFF Challenges

Under or Over Extrusion

- Extruder wells not optimized
- Clogged



Delamination

- Caused by layers shrinking causing cracking
- Material dependent



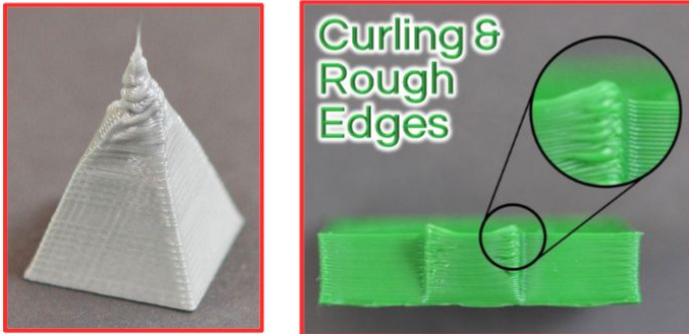
Warping

- Caused by the build plate being too cold and the stress from the cooling peeling the part up



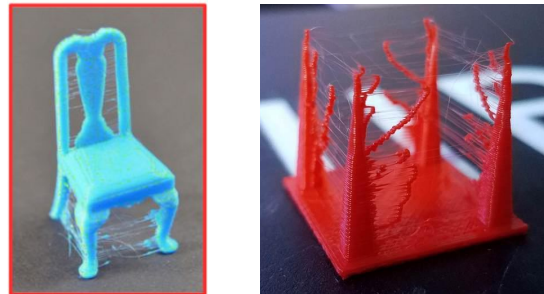
Overheating

- Material doesn't cool down enough before next layer and it starts to



Retraction

- Filament is too hot or extruder movement is moving too fast
- Results in stringing



Elephant Footing

- Caused by the build plate being too hot and melting the base layers



Questions?